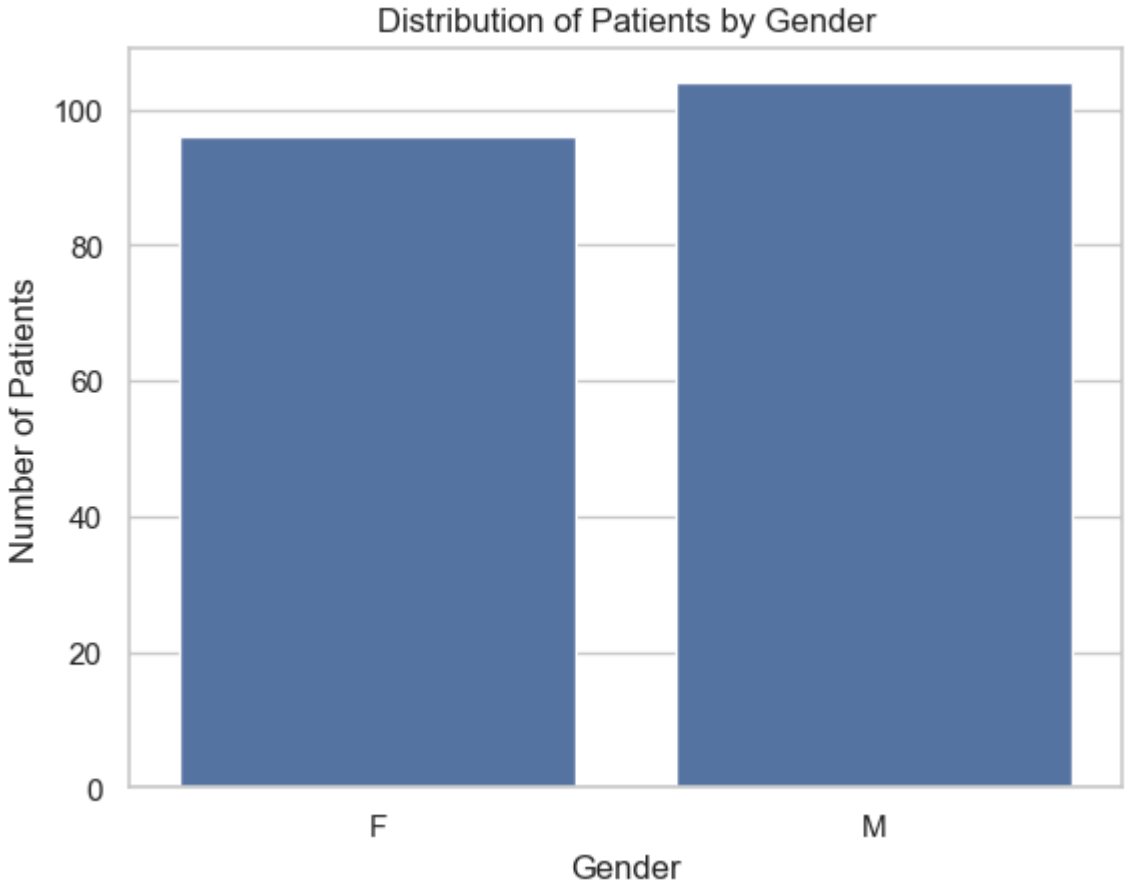
In [15]:
```
import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

sns.set_theme(style="whitegrid")

df = pd.read_csv("drug200.csv")
```

In [16]:
```
sns.countplot(x="Sex", data=df)
plt.title("Distribution of Patients by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Patients")
plt.show()
```



In [17]:
```
print("\nDrug prescription count:")
print(df["Drug"].value_counts())

print("\nDrug with highest prescriptions:")
print(df["Drug"].value_counts().idxmax())
```

```
Drug prescription count:
Drug
drugY    91
drugX    54
drugA    23
drugC    16
drugB    16
Name: count, dtype: int64

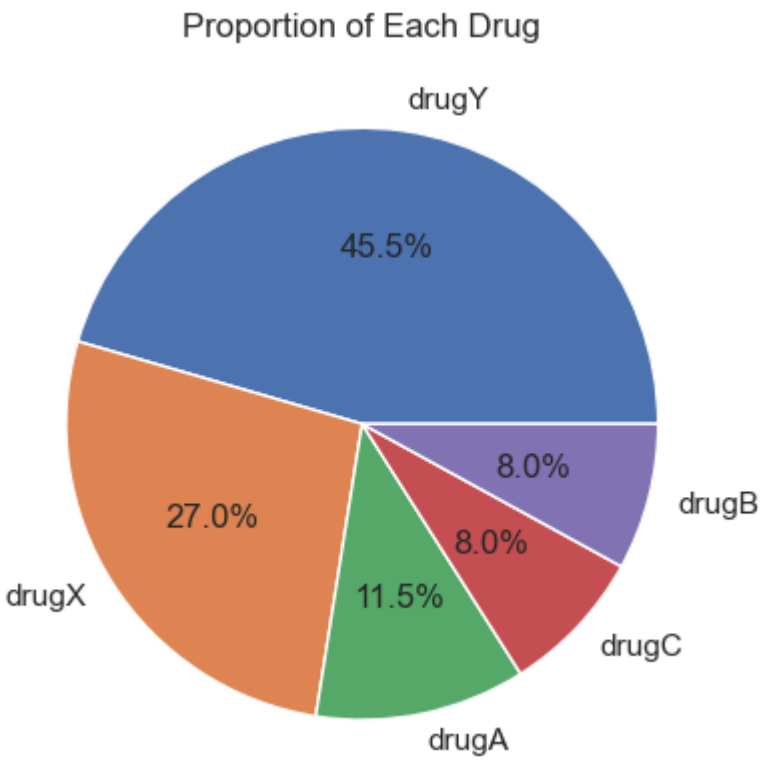
Drug with highest prescriptions:
drugY
```

In [18]:
```
drug_proportion = df["Drug"].value_counts(normalize=True)

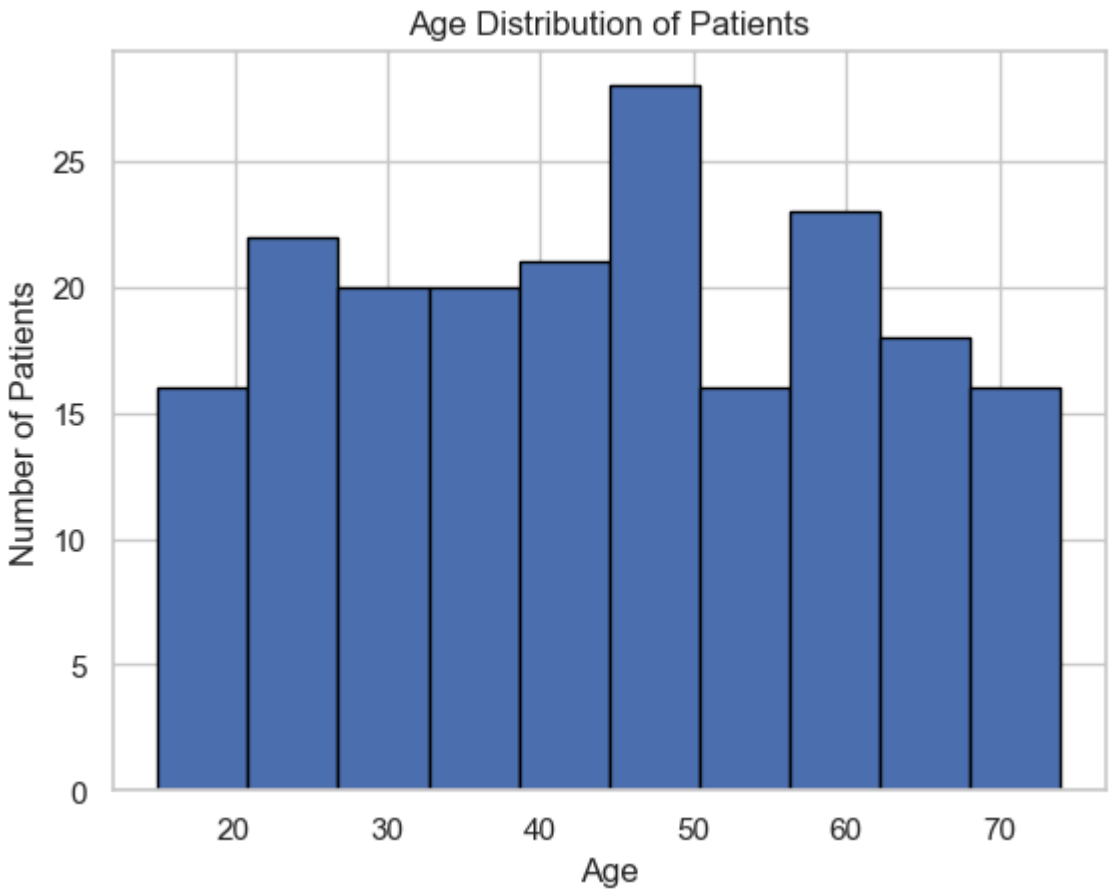
print("\nProportion of each drug:")
print(drug_proportion)

drug_proportion.plot(kind="pie", autopct="%1.1f%%")
plt.title("Proportion of Each Drug")
plt.ylabel("")
plt.show()
```

```
Proportion of each drug:
Drug
drugY    0.455
drugX    0.270
drugA    0.115
drugC    0.080
drugB    0.080
Name: proportion, dtype: float64
```

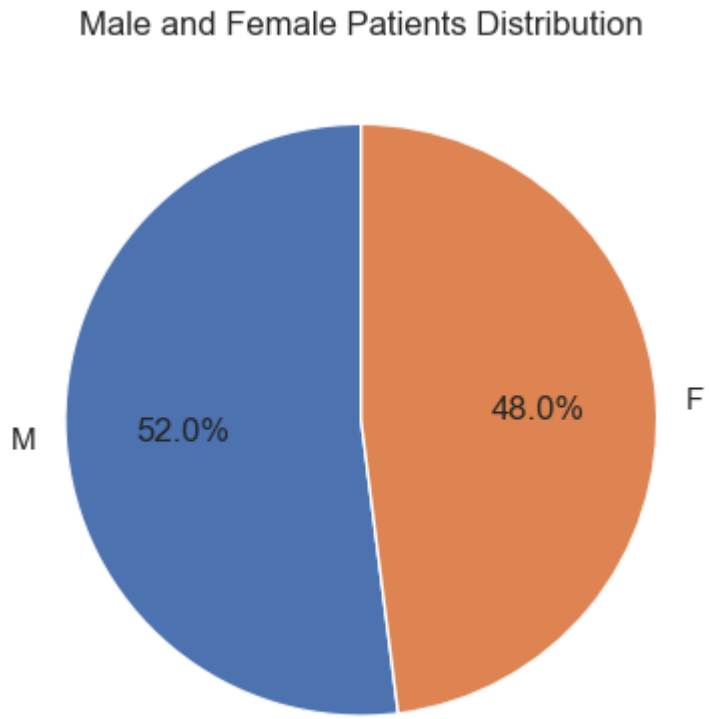


In [19]:
```
plt.hist(df["Age"], bins=10, edgecolor="black")
plt.title("Age Distribution of Patients")
plt.xlabel("Age")
plt.ylabel("Number of Patients")
plt.show()
```

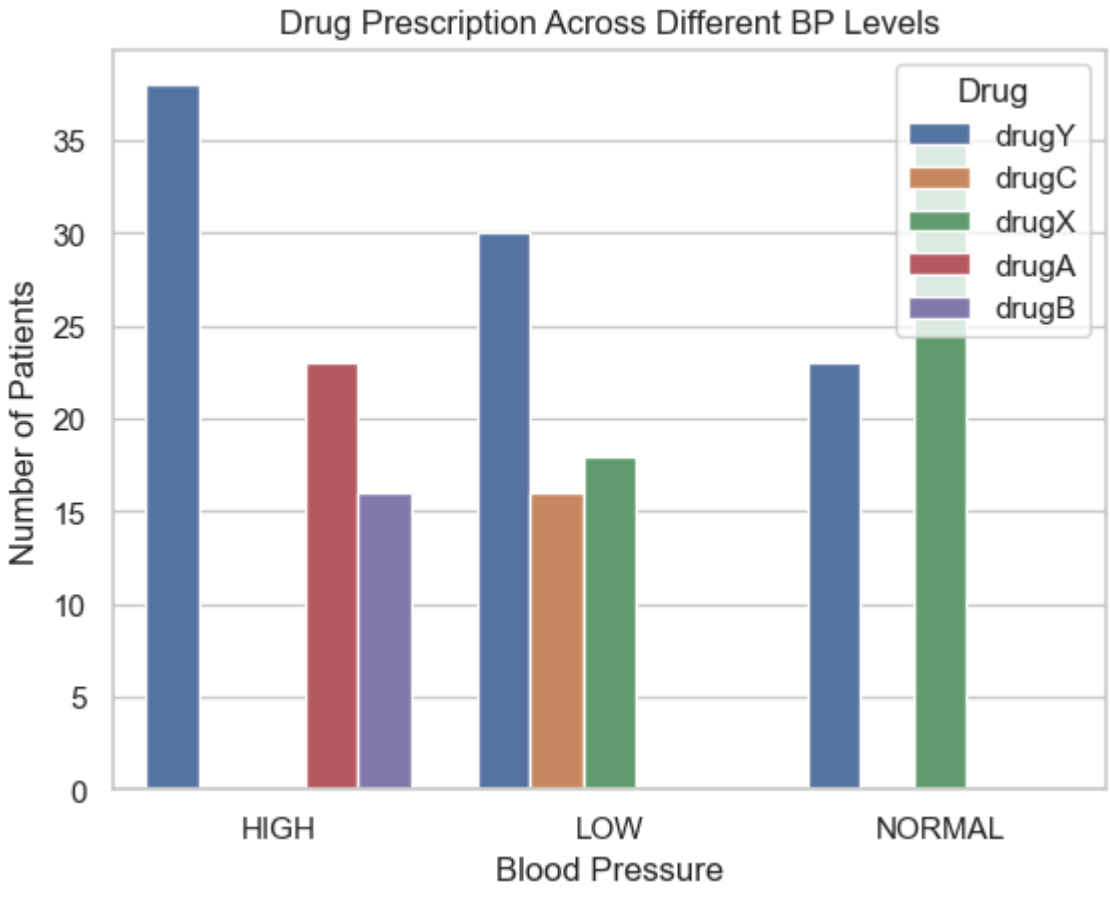


In [20]:
```
gender_count = df["Sex"].value_counts()

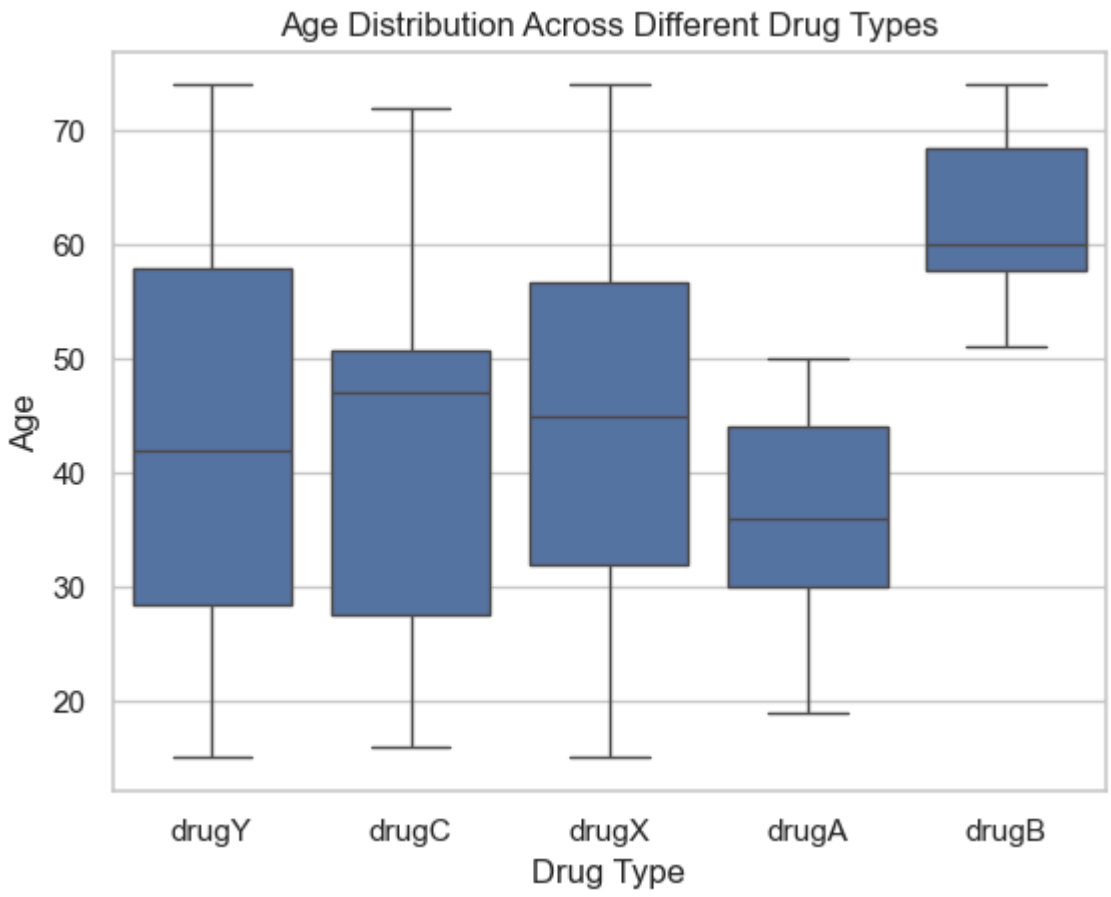
gender_count.plot(
    kind="pie",
    autopct="%1.1f%%",
    startangle=90
)
plt.title("Male and Female Patients Distribution")
plt.ylabel("")
plt.show()
```



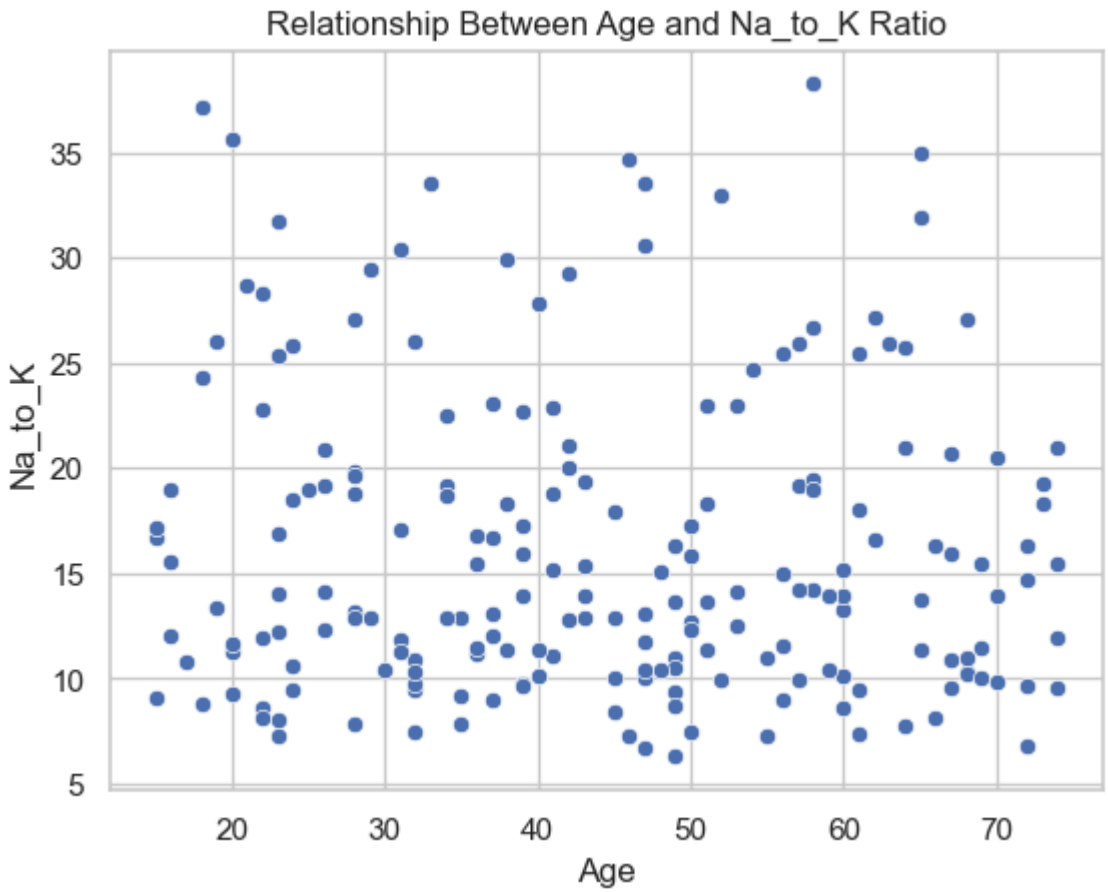
In [21]:
```
sns.countplot(x="BP", hue="Drug", data=df)
plt.title("Drug Prescription Across Different BP Levels")
plt.xlabel("Blood Pressure")
plt.ylabel("Number of Patients")
plt.show()
```



In [22]:
```
sns.boxplot(x="Drug", y="Age", data=df)
plt.title("Age Distribution Across Different Drug Types")
plt.xlabel("Drug Type")
plt.ylabel("Age")
plt.show()
```



In [23]:
```
sns.scatterplot(x="Age", y="Na_to_K", data=df)
plt.title("Relationship Between Age and Na_to_K Ratio")
plt.xlabel("Age")
plt.ylabel("Na_to_K")
plt.show()
```


Relationship Between Age and Na_to_K Ratio
Na_to_K
Age
5
10
15
20
25
30
35
20
30
40
50
60
70

In [ ]: